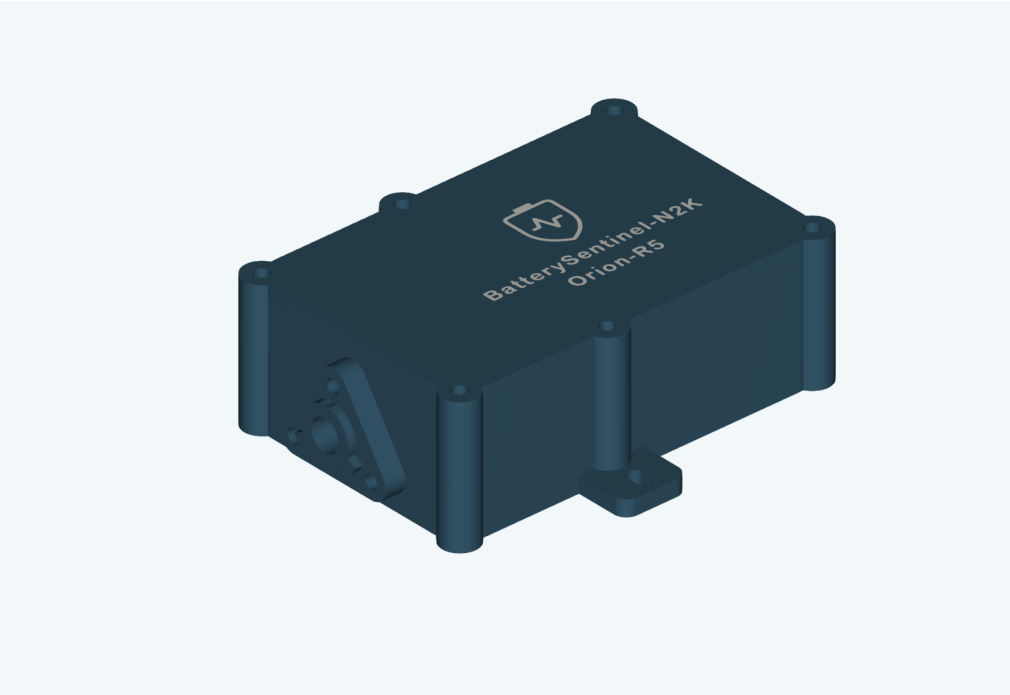
Manufacturing and TPU Air

V2.0 / P2S / 0.4 MM NOZZLE / CALIBRATION FIRST



Actual V2 CAD: extended cable end, three-screw cap and original flush two-colour lid artwork.

Material: **SainSmart Flex TPU Air / lightweight foaming TPU, advertised 77A-93A**. PETG carries the structural loads. The seal and clamp are independent. The file set includes production parts, nine alternative sleeves, fit fixtures, a flow coupon and a separate hardness comparison pad.

Target hardness is not a temperature promise

The Shore ranges in this manual are engineering trial targets. They are not measured results or a supplier-certified temperature-to-hardness table. Accept a setting only after dimensional, compression, leak and retention checks. Do not transfer Siraya Tech TPU Air flow/hardness tables to SainSmart.

Files are prepared for reslicing after calibration, not direct printing of unverified cached G-code. Preserve the prior project and its user test plates. Select the real nozzle and build plate before every print; nozzle type changes require a new calibration.

Hardness targets by function

01 / USE MODERATE FOAMING FIRST

Part	Target Shore A	First temperature / comparison
Lid gasket	80-85 A	245 C; compare 235 and 255 C
USB gasket	83-88 A	235 C; compare 245 C
Switch gasket	83-88 A	235 C; compare 245 C
Cable sealing sleeve	85-90 A	235 C; compare 245 C
Optional clamp liner	Dense 95A; unfoamed	Use its own manufacturer profile
Cap / clamp / case	PETG; no Shore A target	Use the supplied PETG process with calibrated filament

The larger lid loop benefits from the lowest contact force. Small service seals need enough body to keep their continuous bead shape. The cable sleeve is slightly firmer to retain accurate bores and shoulder contact; it does not provide strain relief. These ranges are a starting design preference, not a validated optimum.

Do not aim for 77A everywhere. Maximum foam may lose dimensional accuracy, recovery or impermeability. If a 90A sample passes all functional tests while a softer sample seeps, choose the passing sample and record the departure from the target. Conversely, an in-range Shore value does not prove a watertight seal.

SainSmart supplies a 210-270 C nozzle range, 40-60 C bed range and moderate-speed guidance. The relationship between temperature and final hardness depends on speed, nozzle residence time, flow, cooling and print structure. The comparison temperatures here are experiments within that range, not certified hardness settings. [1]

TPU Air starting process

02 / SETTINGS SHARED BY THE TEMPERATURE TRIALS

Setting	Initial value / qualification
Nozzle / layer / line width	0.4 mm / 0.20 mm / nominal 0.42 mm; recalibrate for actual nozzle
Temperature jobs	235, 245 and 255 C as separate profiles/jobs
First layer / other nozzle temperature	Same temperature; stabilise before printing
Bed	50 C on the selected compatible plate
Walls / infill / top-bottom	3 walls / 100% / 5 and 5 layers
Wall + solid infill speed	20 mm/s; first layer 15 mm/s
Maximum volumetric speed	2.0 mm3/s ceiling; do not use PETG HF limits
Cooling	30% after initial adhesion; auxiliary fan off
Retraction	0.4 mm at 20 mm/s as a trial; tune for actual feed
Flow ratio	1.00 is an UNCALIBRATED seed; calibrate before production
Feed / drying	External spool, low drag; spool-specific drying instructions
Surface	No ironing; no support scars on functional seal faces

100% infill removes intentionally sparse voids; individual extrusions can still foam. It neither prevents foaming nor guarantees a watertight surface. Avoid low infill or deliberate gaps as a softness control. There is no proven unfoamed skin in a uniformly hot printed seal.

Keep speed and cooling stable while comparing temperatures. Small-layer slowdown and the first-layer speed can change expansion. The supplied trials disable cooling-driven slowdown; inspect the actual preview and test the small service seals at their real speed. Stop a print if feed buckles or repeated underextrusion appears.

Temperature and flow calibration

03 / COMPLETE THIS BEFORE FULL SEAL PRINTS

1 Prepare dry material

Check the spool label and batch. Dry with this material/spool supplier instruction and feed from dry storage. Moisture bubbling is not controlled foaming; do not mistake it for a softer successful print.

2 Establish a baseline

Print flow_wall_coupon at 235 C, 20 mm/s, 0.20 mm layers and seed flow 1.00. This small coupon contains a nominal 0.9 mm wall. Check the slicer actually generates continuous paths at the intended wall width.

3 Correct expansion

Measure several wall positions and inspect the solid regions. If overexpanded, reduce flow progressively. Old flow x target/measured width is a rough first correction only; confirm again. Reject settings with gaps, swollen seams or unstable feed.

4 Repeat at each temperature

Make independent 245 C and 255 C jobs and calibrate flow separately. Record nozzle type, speed, flow and dimensions. Never copy a 50% flow guess or assume one flow ratio fits all temperatures.

5 Measure a comparison pad

With each calibrated setting, print shore_comparison_pad and the face/cable specimens. Use the procedure on the next page. The pad and thin seals can differ in thermal history; qualify the actual seal too.

6 Select by function

For each seal choose the least aggressive foaming that reaches suitable compliance and passes the contact/leak/recovery tests. If targets are missed, adjust in 5 C steps within the supplier range and recalibrate flow. Treat 260-270 C as a later experiment, not the default.

Measuring Shore A

04 / A COMPARISON, NOT A CERTIFIED MATERIAL TEST

Use the supplied **50 x 50 x 6 mm** flat comparison pad, printed with 100% infill and the same material/process as the trial. Bareiss describes flat specimens of at least 6 mm thickness for its Shore instruments. Follow your durometer manual for the presser foot, load, dwell time and test-point spacing. [2]

Allow all samples the same conditioning time and temperature; a practical internal comparison is after 24 hours in the same room. Support the pad flat on a rigid surface. Use a stand if available, approach perpendicular and avoid an edge, embossed mark, seam or visibly defective area. Never measure the 1.8 mm gasket against a hard table and call that the bulk Shore value.

Take at least five spaced readings using the instrument-prescribed dwell. Record all readings, the mean, range, dwell, room temperature and conditioning. Apply the same method throughout. If instrument instructions allow both, record separate 1-second and 15-second values; do not mix them in one average. Printed foam is structured and anisotropic, so these are process-comparison values, not certified ASTM/ISO material properties.

Trial record	Write measured values
Temperature / nozzle type / speed	_____
Flow / fan / batch / drying	_____
Pad size / mass / conditioning	_____
Readings 1 / 2 / 3 / 4 / 5	_____
Mean / min-max / dwell / ambient	_____
Actual gasket contact / leak / recovery	_____

Without a durometer

Record hardness as unmeasured. A finger squeeze or scale fixture can compare force and recovery, but cannot establish Shore A. You can still screen the functional seals; do not label them with a fabricated Shore value.

Print orientation and rigid parts

05 / SEPARATE PETG AND TPU JOBS

Part	Orientation / critical point
Base + entry coupon	Outside floor on plate. Inspect horizontal insert/socket roofs.
Two-colour lid	Exterior face down. Body and markings share one transform; never arrange markings independently.
Service covers	Groove face down, exterior reinforcement up. Inspect the 3.4 mm groove-roof bridges before assembly.
Pressure cap	Flat sealing side down, reinforcement ribs up. Keep washer seats flat.
Upper cable clamp	Flat top down; open saddle upward, support-free.
Face seals	Flat carrier down; rounded bead upward. No brim touching the bead.
Cable sleeves	Cable axis vertical; raised shoulder bead up. No support in the bore.
Shims	Flat. 0.2 mm is one layer; inspect for completeness. Never mix unequal thicknesses.

PETG starting process: 0.20 mm layers, 6 walls, 6 top/bottom layers, 35% gyroid, outer/inner walls 40/80 mm/s and 12 mm3/s volumetric ceiling. The small cap and upper clamp use 100% infill overrides. The two PETG colours must be compatible. The saved prior project is retained; these new, separately named V2 process settings are conservative prototype starts, not an audit of a live slicer profile.

Use the supplied print orientations and scale 100%. Choose the actual build plate and calibrated PETG/TPU filament. A smooth plate with an appropriate release layer may improve contact faces; remove parts after cooling without stretching seals. Do not automatically smooth, decimate or scale all parts to compensate for one bore.

Fit and compression experiments

06 / SELECT A SLEEVE AND CLAMP SETTING

Variable	Alternatives / purpose
Cable bead bore	6.9 / 6.8 / 6.7 mm for the measured 7 mm jacket
Sleeve free length	6.6 / 6.8 / 7.0 mm into a 6.0 mm seat
Axial squeeze	9.1% / 11.8% / 14.3% respectively
Initial sleeve	b6p8_L6p8; compare adjacent alternatives if needed
Face seal	1.8 mm free, 1.4 mm production stop; 22.2% nominal squeeze
Face test fixture	Compare 1.4 and 1.6 mm stops with a straight section cut from the spare face_test_gasket; a flat plate supplies the load
Clamp shims	0.4 mm first, then 0.2 mm, then none only if needed

The outer sleeve body has relief space. Its two internal rounded beads contact the jacket; the annular end bead contacts the pocket shoulder. Inspect both paths. A cable grip test does not establish shoulder sealing, and a watertight sleeve does not establish strain relief.

Test insert seating and cap closure on entry_fit_coupon. It reproduces the actual horizontal print orientation. Do not chase a leak by adding torque after the hard stops meet. Check flow, bead continuity, bore dimensions and jacket ovality first. Complete the final exposure test on the complete empty case, not only the open coupon.

Keep specimens compressed for 24-48 hours. Check load relaxation, cap bowing and seepage; after release, measure recovery at a fixed recorded time. Repeat under representative warmth and after opening/closing cycles. If a part takes excessive permanent set or leaks, its Shore target alone is irrelevant.

Release checklist and sources

07 / EVIDENCE REQUIRED BEFORE BOAT INSTALLATION

1 Geometry and slicing

Confirm validation.json passes and package-validation.json identifies the actual slicer used. Open the prepared project, check object/material mapping, plate bounds, first-layer contact, bridges and every thin seal bead. Re-slice after any calibration change.

2 Material process

Record final per-temperature flow, measured pad hardness or unmeasured status, and actual seal dimensions. No supplier-approved hardness/temperature curve is supplied for this printer/nozzle.

3 Mechanical and water checks

Complete the assembly manual: insert fit, independent clamp retention, jacket inspection, 24-48 h recheck and empty-case leakage before/after cable flexing and service cycles.

4 Fallback if foam fails

If water permeates the material or it loses compression, retain the separate clamp/stiff cap arrangement and redesign the seat for a selected conventional silicone/EPDM seal or commercial cable gland. Do not claim waterproofness from a closed STL.

Sources checked 27 September 2026

[1] SainSmart, Lightweight TPU Filament / Foaming LW-TPU 77A-93A: sainsmart.com/products/lightweight-tpu-filament. Material range and general print guidance; not a validated P2S seal recipe.

[2] Bareiss, Digital Durometer Hardness Tester: bareiss-testing.com/digital-durometer-hardness-tester/. Specimen/instrument context; follow your own instrument instructions.

Geometry and dimensions: sealed-v2/parameters.json, current PCB body model, CAD export and digital validation. Shore ranges, comparison temperatures and acceptance sequence are project engineering proposals requiring physical validation.